

Time Series Analysis

Persistence, Trends, and Spurious Regression

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What changes when data are ordered in time?

We leave the cross-section of the basics chapter: instead of many units seen once, we follow **one unit over time** t — GDP, inflation, an exchange rate, a firm's sales.

- Order matters: this period resembles the last. Observations are **dependent**.
- Two consequences run through the whole lecture:

The two problems

- (1) **Serially correlated errors** $\Rightarrow$ the OLS coefficient is still fine, but its **standard errors are wrong**.
- (2) **Trends and persistence** $\Rightarrow$ a regression of one wandering series on another can be **completely spurious**.

Roadmap

- ➊ **Serial correlation** — and honest standard errors.
- ➋ **Dynamics** — distributed lags and autoregression.
- ➌ **Stationarity & spurious regression** — the trap, shown by simulation.
- ➍ A short **Python** session, and the bridge to **panel data**.

Textbook, with replication code in R / Python / Stata: Békés and Kézdi (2021).

Section 1

Serial correlation

Serially correlated errors

In a time series the shock in one period spills into the next:

$$y_t = \alpha + \beta x_t + \varepsilon_t, \quad \text{Cov}(\varepsilon_t, \varepsilon_{t-s}) \neq 0.$$

- Like heteroskedasticity, this leaves OLS **unbiased and consistent** (if x is exogenous) — but the usual **standard errors are wrong**, typically **too small**.
- So t -statistics are inflated and conventional tests **over-reject the null**.
- Cause: inertia, slow adjustment, omitted dynamics — last period's shock is still here today.

This is the same disease that clustering cures in the panel chapter.

The fix: HAC (Newey–West) standard errors

You met this idea in the basics chapter; for a time series it is the **default**.

Heteroskedasticity- and autocorrelation-consistent (HAC) SE

Newey–West standard errors (Newey and West, 1987) widen the usual SE to let errors correlate across nearby periods, up to a chosen lag window.

- Keep the OLS point estimate; **only the standard error changes** (a sandwich, as in the panel chapter).
- Detection is secondary: inspect the residuals or apply the Breusch–Godfrey test (basics chapter); the correction is identical either way.
- As a default, report HAC standard errors for any **stationary** time-series regression. A non-stationary, wandering series is a different problem (later) that HAC does **not** cure.

Section 2

Dynamics: lags

Distributed lags: the effect of x over time

HAC fixed the *symptom* — wrong standard errors. The deeper cause is often **missing dynamics**: x acts not at once but **over several periods** (a price change, a policy, a demand shock):

$$y_t = \alpha + \beta_0 x_t + \beta_1 x_{t-1} + \cdots + \beta_k x_{t-k} + \varepsilon_t.$$

- β_0 — the **immediate** (same-period) effect.
- β_1 — the effect **one period later**, and so on.
- A **distributed-lag** model traces the *dynamic response* of y to a change in x .

The long-run (cumulative) effect

The full effect of a **permanent** one-unit rise in x is the **sum** of the lag coefficients:

$$\beta_{LR} = \beta_0 + \beta_1 + \cdots + \beta_k.$$

- It is the **total** response of y to a permanent one-unit change in x .
- Its **standard error** is not printed directly — recover it with a one-line reparametrization (regress on the Δx terms) or ask the software for the linear combination.
- This long-run logic returns the moment we put a lagged y on the right.

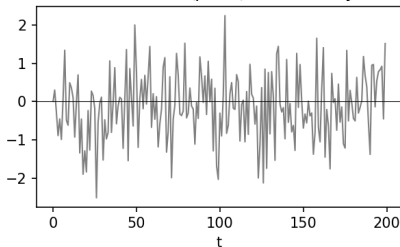
Autoregression: a series explained by its own past

Some series are driven mainly by their **own history** (asset prices, exchange rates), with no obvious external x :

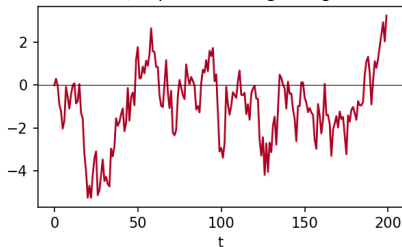
$$\text{AR}(1) : y_t = \alpha + \rho y_{t-1} + \varepsilon_t.$$

- ρ measures **persistence**: how much of today carries into tomorrow.
- $\rho \approx 0 \rightarrow$ looks like noise; ρ close to 1 $\rightarrow$ long swings, slow reversion.

White noise ($\rho = 0$): no memory



AR(1), $\rho = 0.85$: long swings



Persistence is not residual autocorrelation

A common confusion — keep the two apart:

Two different things

Persistence: the *dependent variable* depends on its own past (ρy_{t-1} on the right) — a feature you *model*.

Residual autocorrelation: the *errors* are correlated (§1) — a problem you *fix* with HAC standard errors.

- You can have either one without the other.
- Adding lags of y often **removes** residual autocorrelation: the omitted dynamics had been absorbed into the error.
- Mixing own past *and* an external x gives the **ADL** (autoregressive distributed-lag) model.

Section 3

Stationarity

Stationarity: what it means, why it matters

Standard OLS inference relies on the series being **stationary and weakly dependent** — its behaviour does not drift over time, and the memory of past shocks **fades**.

(Weak) stationarity

Constant mean $\mathbb{E}[y_t] = \mu$, constant variance $\text{Var}(y_t) = \sigma^2$, and an autocovariance that depends only on the *gap* s , not on the date t .

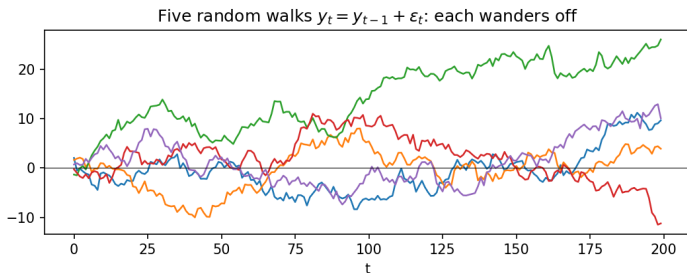
- **Non-stationary** = the mean, variance, or autocorrelation changes over time.
- Usual culprits: a **trend** (drifts up or down) and **seasonality** (regular cycles).
- Diagnose it by **inspecting the series** (trend? seasonal cycle?) and, formally, with the **unit-root tests** below.

Random walks and unit roots

The key non-stationary case is the **random walk** — an AR(1) with $\rho = 1$:

$$y_t = y_{t-1} + \varepsilon_t.$$

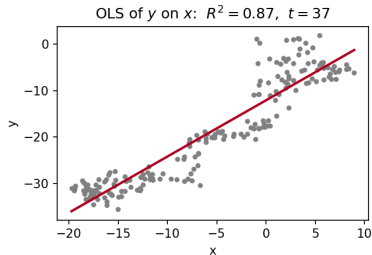
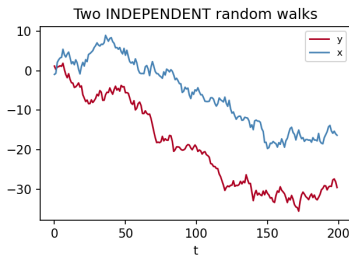
- Each shock is **permanent**: it never fades, and the series **wanders** with no mean to return to. Its variance **grows with time** (a “unit root”: the AR coefficient is 1).
- Contrast: an AR(1) with $\rho < 1$ is **stationary** — shocks die out, the series reverts.



The trap: spurious regression

Granger & Newbold (1974)

Regress one wandering series on another, **unrelated** one: two **independent** random walks routinely give a **huge** R^2 and a **very significant** t — a pure illusion.



The series are unrelated; OLS mistakes their **persistent wandering** — each drifts far in-sample by chance — for a relationship. **On non-stationary data, conventional t and R^2 inference is invalid.**

Detecting and treating non-stationarity

Unit-root and stationarity tests (know the null of each)

Augmented Dickey–Fuller (ADF): H_0 = a unit root (non-stationary); rejection favours stationarity.

KPSS: H_0 = stationarity — the complementary null.

Phillips–Perron: an ADF variant robust to autocorrelation and heteroskedasticity.

If a series is non-stationary, **transform** it:

- take **logs** / **growth rates**, or **first-difference** $\Delta y_t = y_t - y_{t-1}$ to remove a trend; add a **trend term** t or **seasonal dummies** as required.
- Differencing isolates **short-run** effects (changes, not levels); estimate with **HAC** standard errors.

Section 4

Applied

Time series in Python (statsmodels)

```
import statsmodels.formula.api as smf
df["dy"]      = df["y"].diff()          # first difference
df["dx"]      = df["x"].diff()
df["dx_l1"]   = df["dx"].shift(1)       # one lag
df["dy_l1"]   = df["dy"].shift(1)
# ADL in differences, Newey-West (HAC) SE
m = smf.ols("dy ~ dx + dx_l1 + dy_l1", data=df).fit(
    cov_type="HAC", cov_kwds={"maxlags": 4})
m.summary()
```

- `.diff()` first-differences; `.shift(1)` builds a lag.
- `cov_type="HAC"` → **Newey–West** SE; set `maxlags` $\approx T^{1/4}$.
- Unit-root check: `sm.tsa.adfuller(df["y"])`.
- Full case-study code:
github.com/gabors-data-analysis/da_case_studies.

Takeaways, and the bridge to panel

- Time order brings **dependence**: errors correlate ($\rightarrow$ **HAC** standard errors) and series **persist**.
- **Dynamics**: distributed lags trace x 's effect over time; the AR model captures a series' own inertia (and its long-run effect).
- **Stationarity is essential**: on trending or random-walk data, OLS inference is **spurious** — inspect, transform, and difference as needed.
- Beyond this introduction: ARIMA modelling, vector autoregressions (VAR), and cointegration — developed in the second-year sequence.

Next: panel data

Add a **cross-section** to the time dimension — many units, each over time. Serial correlation $\rightarrow$ **cluster by unit**; trends $\rightarrow$ **time effects**; a lagged $y_{t-1} \rightarrow$ **dynamic panels**.

References

Textbook with replication code (R / Python / Stata): Békés and Kézdi (2021) — github.com/gabors-data-analysis/da_case_studies.

Békés, Gábor, and Gábor Kézdi. 2021. *Data Analysis for Business, Economics, and Policy*. Cambridge University Press.

Granger, C. W. J., and P. Newbold. 1974. "Spurious Regressions in Econometrics." *Journal of Econometrics* 2 (2): 111–20. [https://doi.org/10.1016/0304-4076\(74\)90034-7](https://doi.org/10.1016/0304-4076(74)90034-7).

Newey, Whitney K., and Kenneth D. West. 1987. "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica* 55 (3): 703–8. <https://doi.org/10.2307/1913610>.